33. Anycast, GeoDNS, and Global Traffic

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Phase: Phase 10 — Load balancing and proxies
Estimated time: 1 day
Slug: `33-anycast-dns-cdn`

Learning objectives

- Explain anycast routing idea
- Contrast with unicast
- Relate to DNS/CDN POPs
- Debug geo differences

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

Anycast: same IP announced from many places; users hit nearby POP.

```
dig +short one.one.one.one
ping -c 2 1.1.1.1
curl -s https://cloudflare.com/cdn-cgi/trace | head
curl -sI https://example.com | grep -i cf-ray
```

Different resolvers/regions may see different answers (GeoDNS).

Common mistakes

- Expecting one physical server behind 1.1.1.1
- Debugging only from one region
- Ignoring resolver location effects

Exercises

Work in `exercises/33-anycast-dns-cdn/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/33-anycast-dns-cdn/` until after an honest attempt.

Self-check

1. Anycast vs unicast?
2. Why region matters?

3. CDN POP meaning?

Further reading

- https://www.cloudflare.com/learning/cdn/glossary/anycast-network/